

SRV HW4

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Question 1

(i)

$$g(x) = \int_0^{b(x)} f(t) dt \quad (1)$$

$$\frac{d}{dx} g(x) = \frac{d}{dx} \int_0^{b(x)} f(t) dt \quad (2)$$

We can define $F(t) = \int f(t) dt$. Thus for arbitrary bounds $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}$. We have the fundamental theorem of calculus

$$\int_{\alpha}^{\beta} f(t) dt = F(\beta) - F(\alpha) \quad (3)$$

Which gives

$$\frac{dg}{dx} = \frac{d}{dx} [F(b(x)) - F(0)] \quad (4)$$

By chain rule we have

$$\frac{dg}{dx} = f(b(x)) b'(x) \quad (5)$$

(ii)

Similarly we have

$$\frac{dh}{dx} = \frac{d}{dx} [F(0) - F(a(x))] \quad (6)$$

$$\frac{dh}{dx} = -f(a(x)) a'(x) \quad (7)$$

(iii) We know that

$$\int_{\gamma}^{\delta} f(t) dt = \int_{\gamma}^0 f(t) dt + \int_0^{\delta} f(t) dt \quad (8)$$

Thus we can use part (i) and (ii) to get

$$\frac{dr}{dx} = f(b(x)) b'(x) - f(a(x)) a'(x) \quad (9)$$

(iii) We see here we have the same case as (iii) with $f(t) = \sin(t^3 - 3t^2 + 3t)$, $b(x) = 1 + x^2$ and $a(x) = 1 - x^2$. So we have

$$\frac{dr}{dx} \rightarrow \frac{dq}{dx} = 2x \left[\sin \left((1 + x^2)^3 - 3(1 + x^2)^2 + 3(1 + x^2) \right) + \sin \left((1 - x^2)^3 - 3(1 - x^2)^2 + 3(1 - x^2) \right) \right] \quad (10)$$

$$\frac{dq}{dx} = 2x \left[\sin(x^6 + 1) + \sin(-x^6 + 6x^2 - 5) \right] \quad (11)$$

Question 2 (i) Here we see we can apply Fubini's theorem as we have a continuous function on a closed region.

Let F^i denote the integral of $f(x, y, z)$ wrt i , where $i \in [x, y, z]$ at its limit points i.e

$$F^x = \int_0^1 f(x, y, z) dx = F(1, y, z) - F(0, y, z) \quad (12)$$

In this notation we are given in the question to show that

$$F^z F^y F^x = F^x F^z F^y \quad (13)$$

Fubini's theorem says we can interchange these ie $F^x F^y = F^y F^x$. Then by Fubini on the RHS we can swap F^x and F^z , then F^x with F^y .

$$F^z F^y F^x = F^z F^y F^x \quad (14)$$

(ii) Given

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{otherwise} \end{cases} \quad (15)$$

We see $f(x, y)$ is well defined $\forall (x, y) \neq (0, 0)$ as it comprises of polynomials with no division by zero. Thus we only need to examine (x, y) near $(0, 0)$. As our $f(x, y) \equiv 0$ at $(0, 0)$ It should approach 0 near $(0, 0)$. For $(x, y) \neq (0, 0)$ we have

$$|f(x, y)| = \left| \frac{xy(x^2 - y^2)^2}{x^2 + y^2} \right| \quad (16)$$

Since $|x^2 - y^2| \leq x^2 + y^2$ gives $(x^2 - y^2)^2 \leq (x^2 + y^2)^2$. We have

$$|f(x, y)| = \left| \frac{xy(x^2 - y^2)^2}{x^2 + y^2} \right| \leq \frac{|x||y|(x^2 + y^2)^2}{x^2 + y^2} \quad (17)$$

So we have $|f(x, y)| \leq |x||y|(x^2 + y^2)$. Since Euclid was clever and gave us $\|(x, y)\| = \sqrt{x^2 + y^2}$. Then we have that $|x| \leq \|(x, y)\|$ and $|y| \leq \|(x, y)\|$. So we can use the fact that $x^2 + y^2 = \|(x, y)\|^2$

$$|f(x, y)| \leq \frac{|x||y|(x^2 + y^2)^2}{x^2 + y^2} \leq \|(x, y)\| \times \|(x, y)\| \times \|(x, y)\|^2 = \|(x, y)\|^4 \quad (18)$$

So we have as for (x, y) near $(0, 0)$ that $\|f(x, y)\| \leq \|(x, y)\|^4$. Thus we can say that

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0 \equiv f(0, 0) \quad (19)$$

We have that $f(x, y)$ is continuous on $[0, 1]^2$.

(iii)

$$\int_{[0, 1]^2} \frac{xy(x^2 - y^2)^2}{x^2 + y^2} dA = \int_0^1 \int_0^1 \frac{xy(x^2 - y^2)^2}{x^2 + y^2} dx dy \quad (20)$$

Taking the integral wrt x evaluating then taking wrt y :

$$y \int_0^1 \frac{(x^5 - 2x^3y^2 + xy^4)}{x^2 + y^2} dx \quad (21)$$

Let $u = x^2 + y^2$ then $du = 2xdx$ and $x^4 = (u - y^2)^2$.

$$= \frac{1}{2} \int_{x=0}^{x=1} \frac{u^2 - 4y^2u + 4y^4}{u} du \quad (22)$$

$$= \int_{x=0}^{x=1} u + \frac{4y^4}{u} - 4y^2 du \quad (23)$$

$$= 2y^4 \ln(u) + \frac{u^2}{2} - 2y^2 u \quad (24)$$

$$= 2y^4 \ln(x^2 + y^2) + \frac{y(x^2 + y^2)^2}{4} - 2y^2(x^2 + y^2) \quad (25)$$

I forgot a factor of y so we bring him back and also evaluate at our bounds ($x = 0$ to $x = 1$). So 23 gives 26

$$= 2y^5 \ln(y^2 + 1) - 2y^5 \ln(y^2) - \frac{3}{2}y^3 + \frac{y}{4} \quad (26)$$

Thus we are left with

$$\int_0^1 2y^5 \ln(y^2 + 1) - 2y^5 \ln(y^2) - \frac{3}{2}y^3 + \frac{y}{4} dy \quad (27)$$

We can use the given result with a factor of 2 out the front. This gives us

$$2 \left(\frac{\ln(2)}{3} - \frac{1}{12} \right) + \int_0^1 -\frac{3}{2}y^3 + \frac{y}{4} dy \quad (28)$$

$$\frac{2 \ln(2)}{3} - \frac{1}{6} - \frac{3}{8} + \frac{1}{8} \quad (29)$$

Finally we get

$$\int_{[0,1]^2} \frac{xy(x^2 - y^2)^2}{x^2 + y^2} dA = \frac{1}{12} [8 \ln(2) - 5] \quad (30)$$